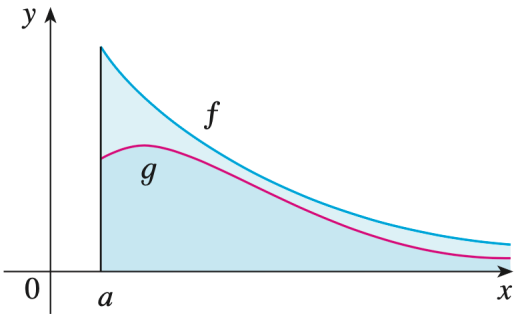
1 The Comparison Test

Sometimes it is impossible to find the exact value of an improper integral and yet it is important to know whether it is convergent or divergent. In such cases the following theorem is useful. We state it below for Type 1 improper integrals, but a similar theorem is also true for Type 2 improper integrals.

Theorem 1.1 (Comparison Test)**.** *Suppose f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.*

(a) If $\int_a^\infty f(x)dx$ is convergent, then $\int_a^\infty g(x)dx$ is convergent.

(b) If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent.



1. The figure on the board shows that graphs of two positive functions f and g defined on the interval $[a,\infty)$.

   (a) Use the figure to explain why the Comparison Theorem seems plausible.

   (b) If $\int_a^\infty g(x)\,dx$ is convergent, must $\int_a^\infty f(x)\,dx$ also be convergent? Argue yes or no using the figure.

   (c) If $\int_a^\infty f(x)\,dx$ is divergent, must $\int_a^\infty g(x)\,dx$ also be divergent? Argue yes or no using the figure.

(d) What if we relax the condition that $g(x) \geq 0$ in the statement of the Comparison Theorem. Do you think it will still be true? Or can you think of an example that shows it is false?

2. Write the statement of the Comparison Test for improper integrals of the form $\int_{-\infty}^{b} f(x)\,dx$, $\int_{-\infty}^{b} g(x)\,dx$.

3. Does $\int_1^\infty \frac{\sin^2(x)}{x^2}\,dx$ converge or diverge?

4. Does $\int_1^\infty \frac{1+e^{-x}}{x}\,dx$ converge or diverge? (*Hint:* Note that $e^{-x} > 0$ for all x.)

5. Show that $\int_0^\infty e^{-x^2}\,dx$ is convergent.

2 Proof of the Comparison Test

Since part (b) of the Comparison Test is the contrapositive of part (a), they are logically equivalent and hence it is sufficient to prove part (a). The proof depends on the following lemma.

Lemma 2.1. *Let $G(t)$ be an increasing function on an interval (a,∞). Assume there exists $M > 0$ such that $G(t) \leq M$ for all $t \in (a,\infty)$. Then $L := \lim_{t\to\infty} G(t)$ exists and $L \leq M$.*

6. We now prove Lemma 2.1.

   Proof. (a) Let $S := \{G(t) : t > a\}$. By assumption, M is an upper bound for S. By completeness of $\mathbb{R}$, $\gamma = \sup S$ exists. Let $\varepsilon > 0$. Show that there exists $t_0 > a$ such that $\gamma - \varepsilon < G(t_0)$.

   (b) Now show that

   $$\gamma - \varepsilon < G(t_0) < G(t) \leq \gamma < \gamma + \varepsilon \text{ for } t > t_0.$$

   (c) Show that (b) implies $|G(t) - \gamma| < \varepsilon$ for all $t > t_0$, which is to say that $\lim_{t\to\infty} G(t) = \gamma =: L$.

   (d) Finally, show that $L \leq M$.

   □

7. We now prove part (a) of the Comparison Test.

Proof. Consider the functions $F(t) = \int_a^t f(x)dx$ and $G(t) = \int_a^t g(x)dx$, defined for $t \geq a$.

(a) Show that F and G are both increasing on $[a, \infty)$.

(b) Show that $G(t) \leq F(t)$ for all $t > a$.

(c) By assumption $\int_a^\infty f(x)dx$ converges to some number M. By definition of the improper integral, $M = \lim_{t\to\infty} F(t)$. Show that $F(t) \leq M$ and $G(t) \leq M$ for all $t \geq a$.

(d) Apply Lemma 2.1 to complete the proof.

□